FRDM-MCXW70

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NOTES

1. Unless Otherwise Specified:
   All resistors are in ohms, 5%, 1/8 Watt
   All voltages are DC

2. Interrupted lines coded with the same letter or letter combinations are electrically connected.

3. Device type number is for reference only. The number varies with the manufacturer.

4. Special signal usage:
   _B Denotes - Active-Low Signal
   <> or [] Denotes - Vectored Signals

5. Interpret diagram in accordance with American National Standards Institute specifications, current revision, with the exception of logic block symbology.

Revision History

Rev. Code	Date	By	Description
A	2025-12-17	Shawn Shi	Initial Release

Jumper Setting

REF DES	JUMPER(DEFAULT)	PAGE NAME
J2,J4,J32,JP1	1-2	03. System PWR
J5	OPEN	03. System PWR
J8,J9,J10,J11,JP2,JP3	1-2	04. PWR Management
J13,J14,J15	1-2	06. HMI/Sensor/Memory
JP21,JP22	1-2	07. CAN/LIN
J17	OPEN	07. CAN/LIN
JP5	1-2	08. Arduino/MikroE/JTAG
JP8,JP20	1-2	09. MCU_LINK_USB
JP6,JP7,JP9	OPEN	09. MCU_LINK_USB
JP12	1-2	10. MCU_LINK_DEBUG
JP10,JP11	OPEN	10. MCU_LINK_DEBUG
JP13,JP14,JP15,JP16	1-2	11. MCU_LINK_CURRENT
JP17	1-2	12. MCU_LINK_VCOM_SIO
JP18,JP19	2-3	12. MCU_LINK_VCOM_SIO

DNP Setting

REF DES	ASSY_OPT	PAGE NAME
C15,J3,J6,J7,R1,R2,R5,U4	DNP	03. System PWR
C19,C20,C23	DNP	04. PWR Management
C43,C44,C45,C46,C47,C51,C136, C137,C138,C139	DNP	05. MCXW70
C53,C54,R62,R73,R76,R77,R90, R91,R92,R93	DNP	06. HMI/Sensor/Memory
J33	DNP	07. CAN/LIN
D10,R140,R141,R142,R144,R147, R149,TP9	DNP	08. Arduino/MikroE/JTAG
C94,C95,J29,J30,R169,R181, R182	DNP	09. MCU_LINK_USB
D21,D22,D23,J31,R188	DNP	10. MCU_LINK_DEBUG
R208,R221	DNP	11. MCU_LINK_CURRENT

ICAP Classification: CP: ___ IUO: X PUBI: ___

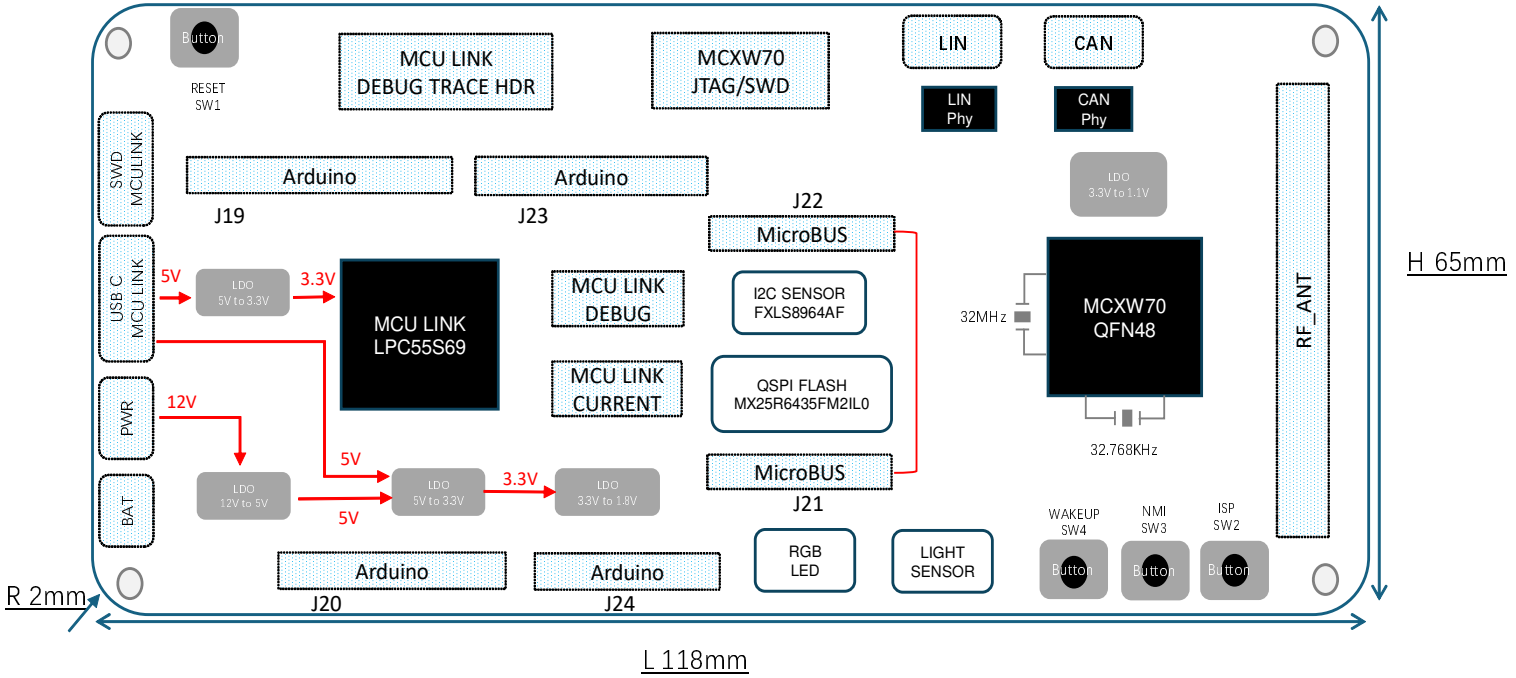
Drawing Title: **FRDM-MCXW70**

Page Title: **Title Page**

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C	SCH-95987 PDF: SPF-95987	A

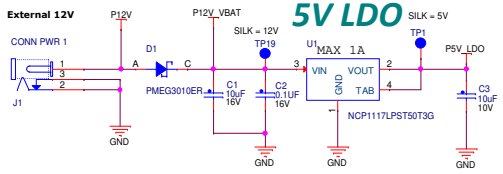
Date: Friday, December 19, 2025 Sheet 1 of 12

FRDM-MCXW70

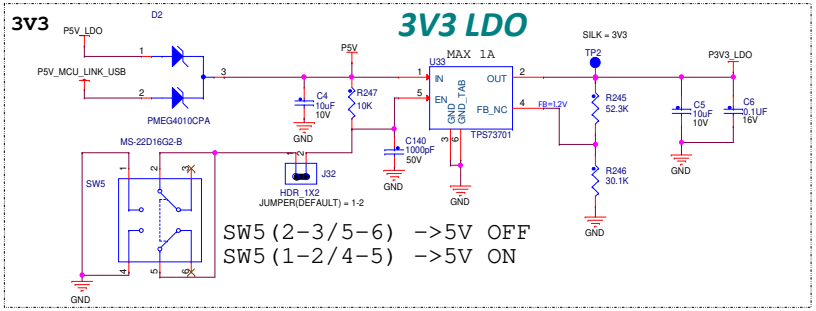


ICAP Classification:	CP: ___ IUO: X PUBI: ___	
Drawing Title:	FRDM-MCXW70	
Page Title:	Block Diagram	
Size: C	Document Number: SCH-95987 PDF: SPF-95987	Rev: A
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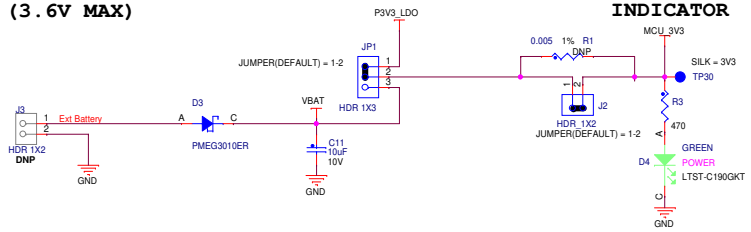
CAN/LIN Power



3V3

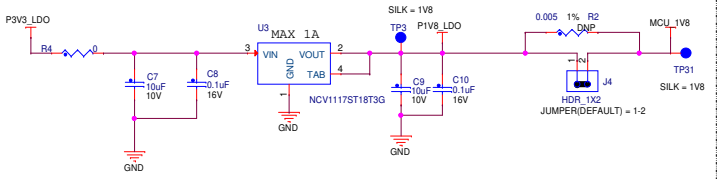


EXT BATTC (3.6V MAX)

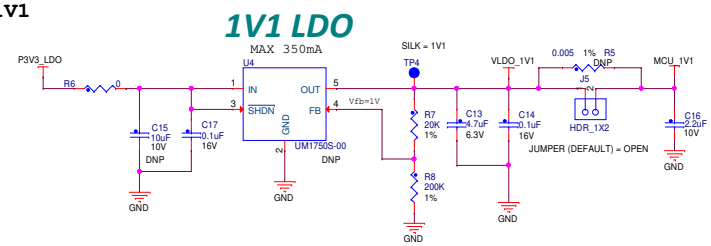


POWER ON INDICATOR

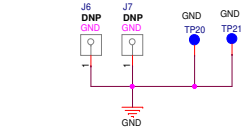
1V8



1V1



GND Test Points



Note:
Two ground test points are placed on the left and right sides of the board to facilitate testing

Mounting Holes



Note:
The mechanical mounting holes are placed on the four vertices of the board

ICAP Classification: CP: ___ IUO: X PUBI: ___

Drawing Title: **FRDM-MCXW70**

Page Title: **System Power**

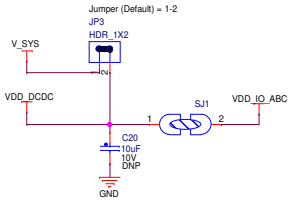
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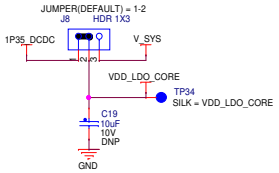
POWER OPTIONS	VDD_IO/VDD_DCDC [1.8V-3.6V]		VDD_LDO_CORE [1.25V-3.6V]	VDD_ANA [1.71V-3.6V]	VDD_VREFH [1.71V-3.6V]	VDD_RF [1.175V-3.6V]
	JP2	JP3	J8	J9	J10	J11
DCDC Mode	1-2	1-2	1-2	1-2	1-2	1-2
LDO Mode	1-2	1-2	2-3	1-2	1-2	2-3

Power Options

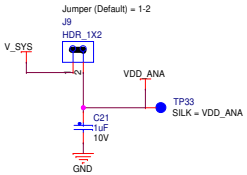
VDD_IO/VDD_DCDC



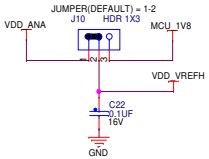
VDD_LDO_CORE



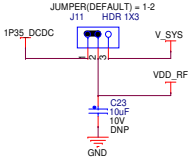
VDD_ANA

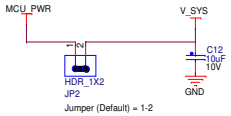


VDD_VREFH



VDD_RF





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MCXW70

U5A
MCXW70ACMFTAR

PTA0/ADC0_A0/TPM1_CLKIN/TRGMUX0_IN1/LPSPI2_SCK/WUU0_P0	5	PTA0	R9	WAKEUP	{6}
PTA1/TDI/ADC0_A1/TPM1_CH0/TRGMUX0_OUT1 /LPSPI2_SIN/WUU0_P1	6	PTA1	R10	TDI	{8,10}
PTA2/TDO/SWO/ADC0_A2/TPM1_CH1/TRGMUX0_OUT2/SWO/LPSPI2_SOUT/WUU0_P6	7	PTA2	R11	TDO_SWO	{8,10}
PTA3/TMS/SWD_DIO/CMP0_OUT/LPUART0_CTS_b/RF_GPO_11/SWD_DIO/WUU0_P17	8	PTA3	R12	TMS_SWDIO	{8,10}
PTA4/TCLK/SWD_CLK/LPUART0_RTS_b/RF_GPO_10/SWD_CLK/TCLK/TP_HCLK/WUU0_P2	9	PTA4	R13	TCK_SWDCLK	{8,10}
PTA5/BOOT_CONFIG/ADC0_A3/CMP0_IN0/ATX0/RF_PLL_TEST_CLK/RF_GPO_9/TRACE_SWO/BOOT_CONFIG/LPSPI2_PCS0/WUU0_P18/RF_XTAL_OUT_ENABLE	10	PTA5	R14	SWO/BOOT_CFG	{6}
PTA16/ADC0_A4/ADC0_A4/LPSPI0_PCS0/EWM0_OUT_b/LPI2C0_SCLS/LPUART0_RX/RF_GPO_8/RF_FPGA_IN0/PWM0_A0/WUU0_P19/RF_NOT_ALLOWED	11	PTA16	R15	LPUART0_RX	{7,12}
PTA17/ADC0_A5/NVM_TM3/ADC0_A5/NVM_TM3/LPSPI0_SIN/EWM0_IN/LPI2C0_SDAS/LPUART0_TX/RF_GPO_7/RF_GPO_8/PWM0_B0/TP_HTRANS/RF_EXT_XTAL_REQUEST/RF_GPO_7/WUU0_P3/RF_NOT_ALLOWED	12	PTA17	R20	LPUART0_TX	{7,12}
PTA18/LPSPI0_SOUT/LPUART0_CTS_b/LPI2C0_SDA/RF_GPO_0/RF_FPGA_IN1/PWM0_A1/LPUART0_RX/SPC0_LPREQ/WUU0_P20	13	PTA18	R21	LPUART0_CTS_b	{12}
PTA19/CMP0_IN1/CMP0_IN1/LPSPI0_SCK/LPUART0_RTS_b/LPI2C0_SCL/RF_GPO_1/RF_FPGA_IN2/PWM0_B1/NVM_SDO/WUU0_P4	14	PTA19	R23	LPUART0_RTS_b	{12}
PTA20/ADC0_A6/CMP0_IN3/ADC0_A6/CMP0_IN3/LPSPI0_PCS2/LPUART0_TX/EWM0_IN/RF_GPO_2/PWM0_A2/TP_LOAD_EN/LPUART0_RTS_b/WUU0_P23	17	PTA20	R25	ADC0_A6	{6}
PTA21/ADC0_A7/CMP0_IN2/ADC0_A7/CMP0_IN2/LPSPI0_PCS3/LPUART0_RX/EWM0_OUT_b/RF_GPO_3/RF_GPO_7/RF_GPO_10/PWM0_B2/TP_HWRITE/LPUART0_CTS_b/WUU0_P5	18	PTA21	R27	INT2_FXLS	{6}
PTB0/ADC0_A9/LPSPI1_PCS0/TPM0_CH3/ADC0_CONV_DATA3/RF_DTEST_2P4GHZ_7/TP_ADDR0/WUU0_P13	46	PTB0	R29	LPSPI1_PCS0	{6}
PTB1/ADC0_A10/LPSPI1_SIN/TPM0_CH4/ADC0_CONV_DATA4/RF_DTEST_2P4GHZ_8/TP_ADDR1/WUU0_P25	47	PTB1	R31	LPSPI1_SIN	{6}
PTB2/ADC0_A11/LPSPI1_SCK/LPUART1_TX/TPM0_CH5/ADC0_CONV_DATA5/RF_DTEST_2P4GHZ_9/TP_ADDR2/WUU0_P26	48	PTB2	R33	LPSPI1_SCK	{6}
PTB3/LPSPI1_SOUT/LPUART1_RX/TPM3_CLKIN/ADC0_CONV_DATA6/RF_DTEST_2P4GHZ_10/PWM0_X0/TP_ADDR3/WUU0_P14	1	PTB3	R35	LPSPI1_SOUT	{6}
PTB4/NVM_TM0/LPSPI1_PCS3/LPUART1_CTS_b/LPI2C1_SDA/TPM3_CH0/TRGMUX0_IN0/ADC0_CONV_DATA7/RF_DTEST_2P4GHZ_11/PWM0_X1/TP_HRESP/WUU0_P15	2	PTB4	R37	LPSPI1_DATA3	{6}
PTB5/NVM_TM1/LPSPI1_PCS2/LPUART1_RTS_b/LPI2C1_SCL/TPM3_CH1/TRGMUX0_OUT0/ADC0_CONV_CLK/RF_DTEST_2P4GHZ_12/PWM0_X2/TP_HREADY/WUU0_P16	3	PTB5	R39	LPSPI1_DATA2	{6}
PTC0/LPSPI1_PCS2/CAN0_TX/TPM4_CLKIN/LPI2C1_SCL/RF_DTEST_2P4GHZ_0/LPSPI2_PCS3/PWM0_A0/WUU0_P7	37	PTC0	R41	LPI2C1_SCL	{6}
PTC1/LPSPI1_PCS3/CAN0_RX/TPM4_CH0/LPI2C1_SDA/RF_DTEST_2P4GHZ_1/PWM0_B0/WUU0_P8	38	PTC1	R42	LPI2C1_SDA	{6}
PTC2/NVM_TM2/LPSPI1_SOUT/LPUART1_RX/LPI2C1_SCLS/TPM4_CH1/RF_DTEST_2P4GHZ_2/TP_DATA0/PWM0_A1/WUU0_P9	39	PTC2	R43	LPUART1_RX	{12}
PTC3/LPSPI1_SCK/LPUART1_TX/LPI2C1_SDAS/TPM0_CLKIN/RF_DTEST_2P4GHZ_3/TP_DATA1/PWM0_B1/WUU0_P24	40	PTC3	R45	LPUART1_TX	{12}
PTC4/LPSPI1_SIN/CAN0_TX/LPI2C1_SCL/TPM0_CH0/ADC0_CONV_DATA0/RF_DTEST_2P4GHZ_4/TP_DATA2/PWM0_A2/WUU0_P10	42	PTC4	R47	CAN0_TX	{7}
PTC5/LPSPI1_PCS0/CAN0_RX/LPI2C1_SDA/TPM0_CH1/ADC0_CONV_DATA1/RF_DTEST_2P4GHZ_5/TP_DATA3/PWM0_B2/WUU0_P22	43	PTC5	R49	CAN0_RX	{7}
PTC6/ADC0_A8/LPSPI1_PCS1/ADC0_CONV_DATA2/RF_DTEST_2P4GHZ_6/WUU0_P11	44	PTC6	R51	INT1_FXLS	{6}
PTC7/TRGMUX0_IN3/TRGMUX0_OUT3/SFA0_CLK/TPM0_CH2/SFA1_CLK/CLKOUT/RF_DTEST_2P4GHZ_13/NMI_b/WUU0_P12/NMI_b/RF_NOT_ALLOWED	45	PTC7	R53	TPM/RGB_B	{6}
PTD0/RESET_b/ADC0_A12	23	PTD0	R54	RST_TGTMCU_B	{6,8,10}
PTD1/ADC0_A13/SPC0_LPREQ/NMI_b/RF_GPO_4/TPM2_CLKIN/RF_GPO_7/RF_FPGA_IN3	24	PTD1	R56	NMI_B	{6}
PTD2/ADC0_A14/LPTMR0_ALT3/TAMPER0/RF_GPO_5/TPM2_CH0/LPTMR1_TRIG_OUT_b/RF_FPGA_IN4	25	PTD2	R58	TPM/RGB_R	{6}
PTD3/ADC0_A15/LPTMR1_ALT3/TAMPER1/RF_GPO_6/TPM2_CH1/TRGMUX0_IN2/LPTMR0_TRIG_OUT_b/RF_FPGA_IN5	26	PTD3	R60	TPM/RGB_G	{6}
PTD4/XTAL32K/LPTMR0_ALT2/TAMPER2/LPSPI2_PCS1	27	XTAL32K			
PTD5/EXTAL32K/LPTMR1_ALT2/LPSPI2_PCS2	28	EXTAL32K			
ANT_2P4GHZ	35	ANT			
XTAL	32	XTAL_RF			
EXTAL	33	EXTAL_RF			
XTAL_OUT	31			XTAL_OUT	{8}

PTA0 R13 0 PTA0_H {8}
PTA1 R15 0 PTA1_H {8}
PTA2 R17 0 PTA2_H {8}
PTA5 R19 0 PTA5_H {8}

PTA18 R22 0 PTA18_H {8}
PTA19 R24 0 PTA19_H {8}
PTA20 R26 0 PTA20_H {8}
PTA21 R28 0 PTA21_H {8}

PTB0 R30 0 PTB0_H {6,8,12}
PTB1 R32 0 PTB1_H {8,12}
PTB2 R34 0 PTB2_H {8,12}
PTB3 R36 0 PTB3_H {8,12}
PTB4 R38 0 PTB4_H {8,12}
PTB5 R40 0 PTB5_H {8,12}

PTC0 R44 0 PTC0_H {8}
PTC1 R46 0 PTC1_H {8}
PTC2 R48 0 PTC2_H {8}
PTC3 R50 0 PTC3_H {8}
PTC6 R52 0 PTC6_H {8}

PTD1 R55 0 PTD1_H {8}
PTD2 R57 0 PTD2_H {8}
PTD3 R59 0 PTD3_H {8}

PTC0 C136 3pF DNP GND
PTC1 C137 3pF DNP GND
PTC2 C138 3pF DNP GND
PTC3 C139 3pF DNP GND

CAD placement:
C24 & C25 close to pin 16;
C26 close to pin41;

SILK = VDD_CORE
TP24
VDD_CORE
C24 4.7uF 6.3V
C25 0.1uF 16V
C26 0.1uF 16V
GND

SILK = VDD_IO_ABC
TP35
VDD_IO_ABC
C29 2.2uF 6.3V
C30 0.1uF 16V
GND

U5B
MCXW70ACMFTAR

VDD_CORE_1	16	VPA_2P4GHZ	36
VDD_CORE_2	41	VREFH	30
VDD_IO_1	4	DCDC_LX	20
VDD_IO_2	21	VDD_FRO	22
VDD_LDO_CORE	15	VSS	49
VDD_ANA	29	VSS_DCDC	19
VDD_RF	34		

SILK = VDD_VREFH
TP32
VDD_VREFH
VDD_PA_2G4
C27 1uF 10V
C28 0.22uF 10V
GND

VDD_DCDC
C34 10uF 10V
C35 0.1uF 16V
GND

VDD_LDO_CORE
C36 1uF 10V
C37 0.1uF 16V
GND

VDD_ANA
C38 1uF 10V
GND

VDD_RF
C39 4.7uF 6.3V
C135 0.1uF 16V
C40 12pF 50V
GND

VDD_FRO
C33 1.5uF 6.3V
GND

DCDC Inductor

L1 2.2uH
LSBHB1608KKT2R2MG
(125C/750MA/345MOHM)

SILK = 1P35_DCDC
TP23
1P35_DCDC
C31 4.7uF 6.3V
C32 0.01UF 50V
GND

C31/C32 place near L1.2 terminal

RF & ANTENNA

SILK = VDD_PA_2G4
TP18
VDD_PA_2G4
C42 12pF 50V
GND
LQG15HZ15NJ02D
L2
L2 15nH

CAD Note: Keep ANT trace short as possible

J12
SLE-100010072-F01
RF_SMA
GND

CAD NOTE:
SMA - Antenna:

Place both capacitor sharing pin 1

GCM1555C1H150GA16D
C48/C43

LQG15HS1N2S02D
L4

C43 15pF DNP

ANT
L3 1.3nH
ANT_RF
L4 1.2nH
RF_50
C48 15pF
ANT_50
L5 0
RF_ANT
ANT1

LQG15HS1N3S02D
L3

C49 3.3PF 50V
GJM1555C1H3R3CB01D
C49

C50 1PF 50V
GJM1555C1H1R0CB01D
C50

C51 2pF DNP
QSCF251Q2R0A1GV001
251R07S2R0BV4S C51

ERJ-2GE0R00X
L5

CAD Note:
50 ohm controlled impedence line

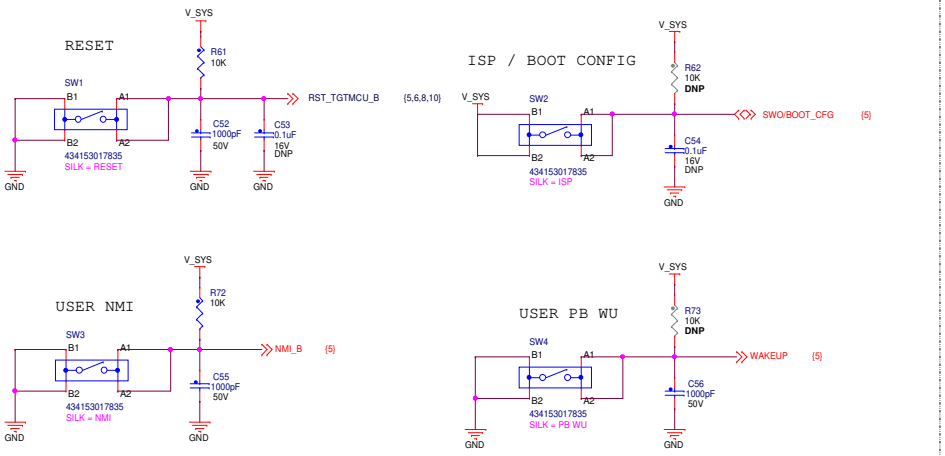
CLOCK SECTION

XTAL_RF
NX1612SA 32 MHz EXS00A-CS15781
Y1
32MHz
32M
R240 0 EXTAL_RF
C44 6.0PF DNP
C45 6.0PF DNP
GND

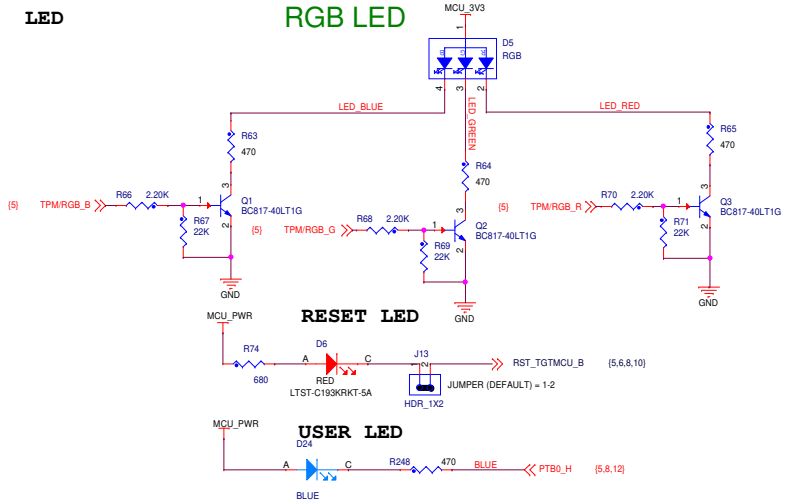
XTAL32K
NX2012SE 32.768kHz EXS00A-MU01517
Y2
32.768 kHz
32K
EXTAL32K
C46 12pF DNP
C47 12pF DNP
GND

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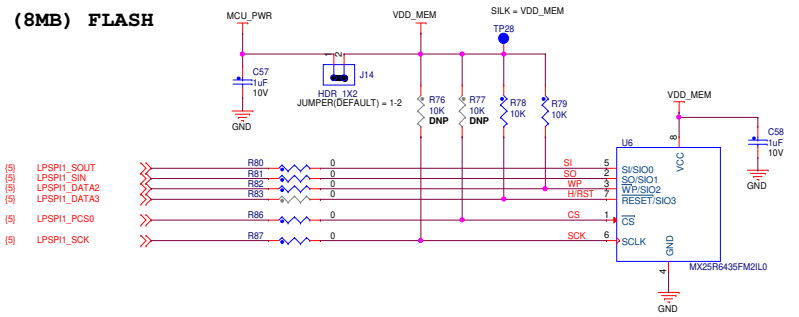
Buttons



LED

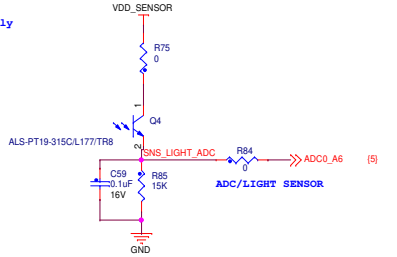


64Mbit (8MB) FLASH

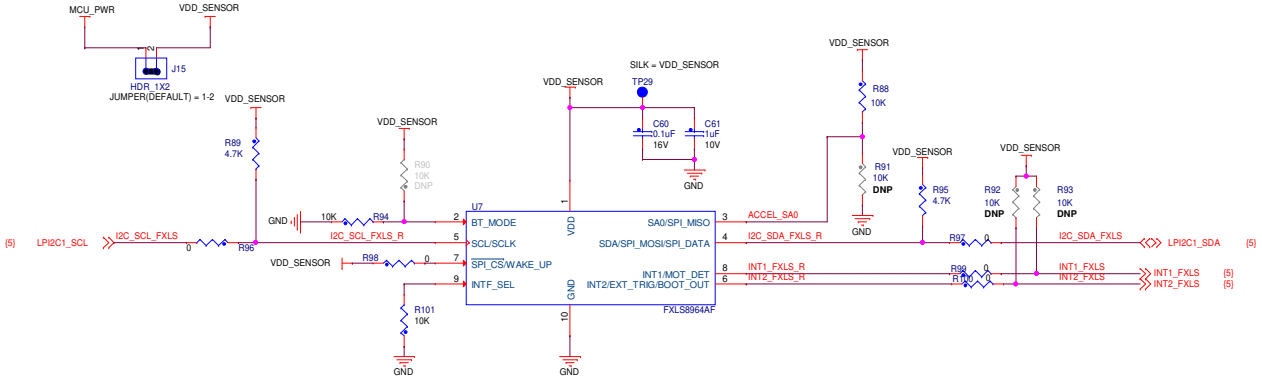


Ambient Light Sensor

Note:
This circuit is working only for 3.3V supply

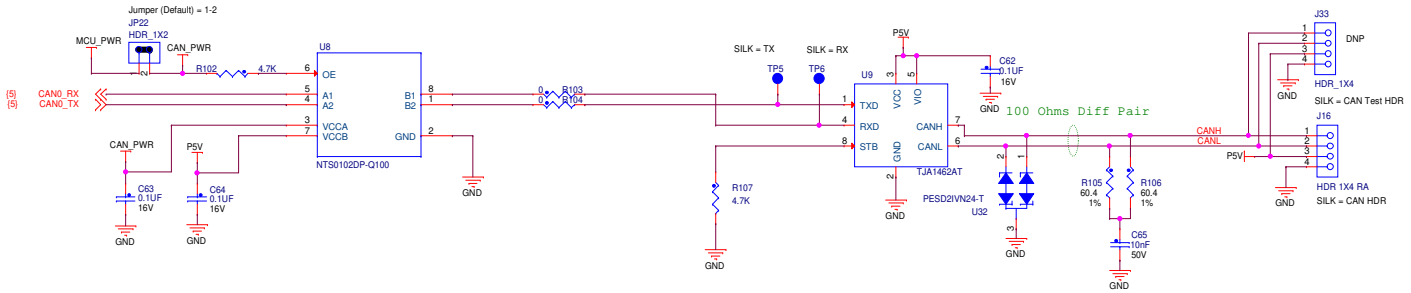


I2C Sensor

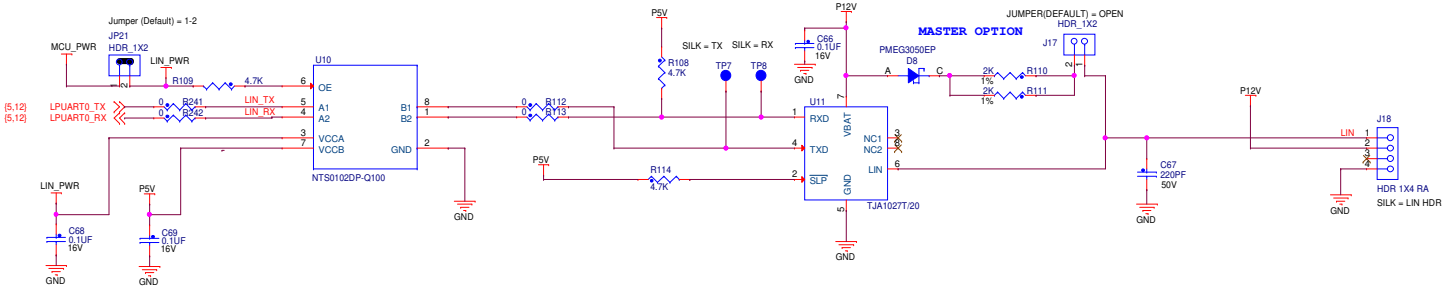


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Drawing Title:	FRDM-MCXW70		
Page Title:	HMI/Sensor/Memory		
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CAN



LIN



ICAP Classification: CP: ___ IUO: X PUBI: __

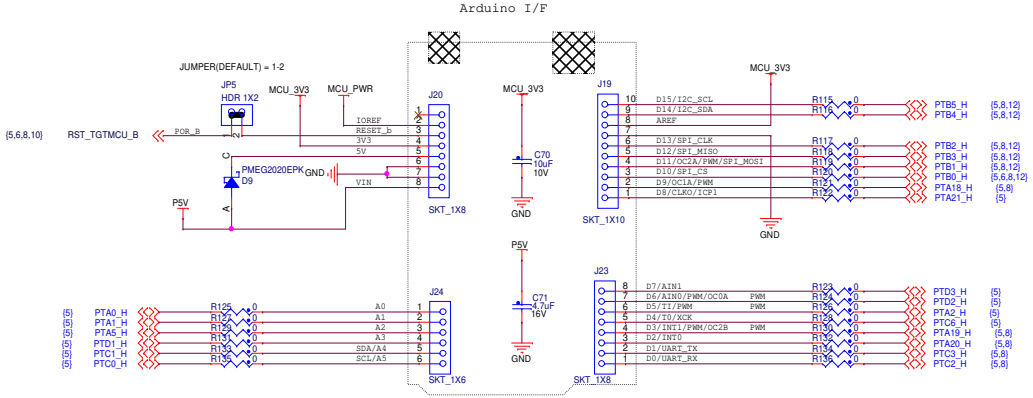
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Arduino



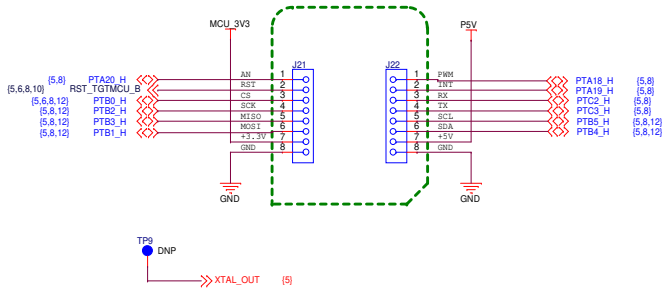
The connectors used are 2.54mm sockets

NOTE: Arduino shield used should not exceed 200 mA power requirements through 3V3.

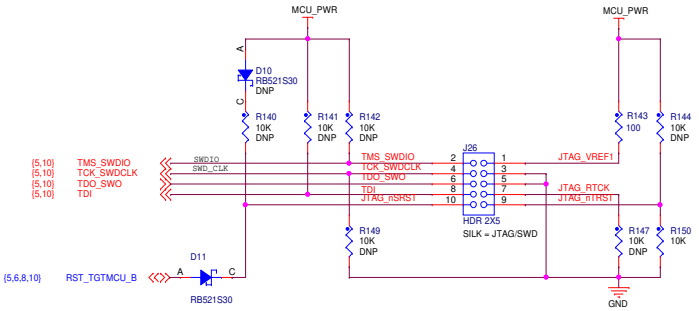
Mikroe

Mikroe Click Connector

(Signals are shared with Arduino.)



JTAG/SWD (DEFAULT)



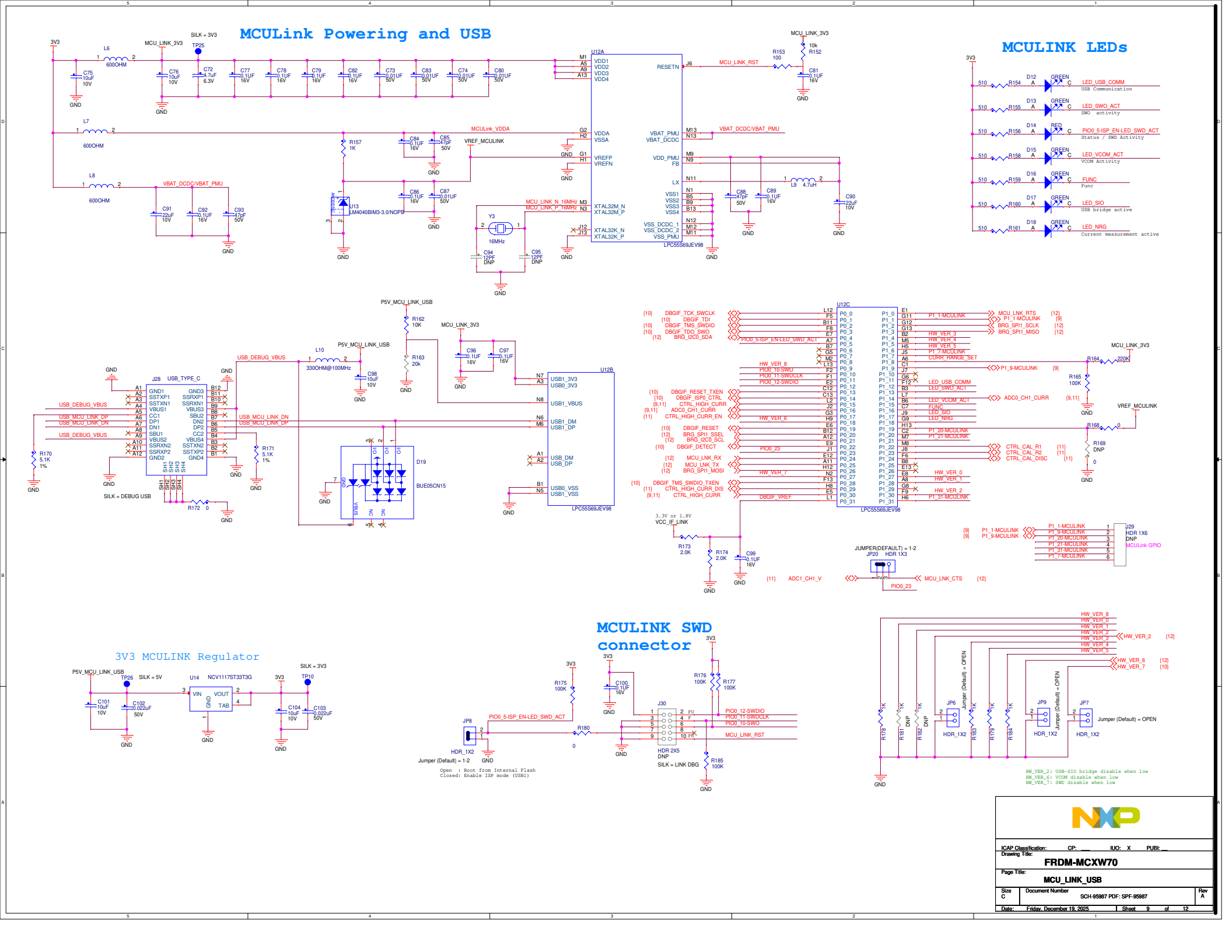
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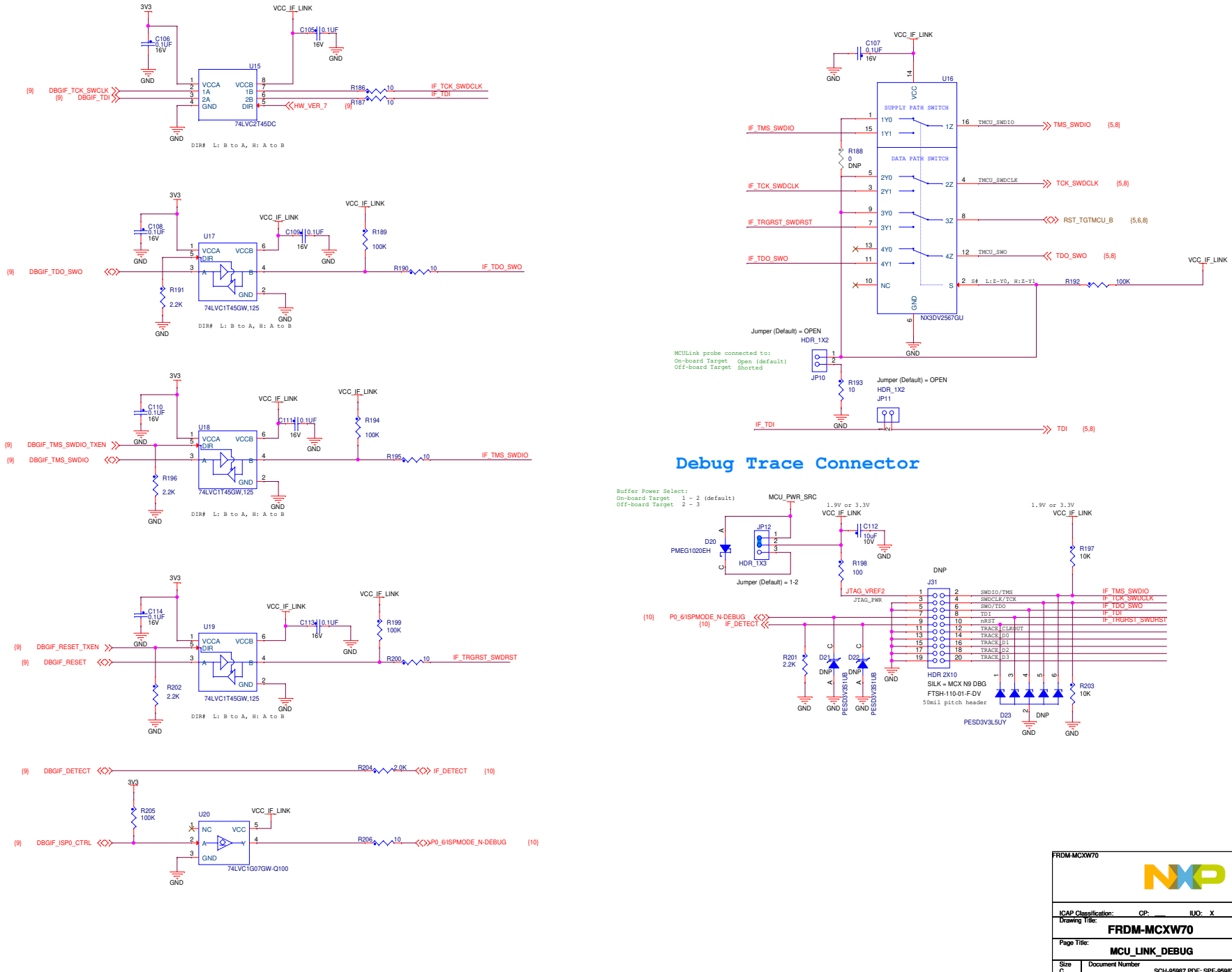
Page Title: **Arduino/MikroE/Debug**

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MCULink Powering and USB
MCULINK LEDs
3V3 Regulator
MCULINK SWD connector
U12A
U12B
U12C
U13 LM4040BIM3-3.0/NOPB
U14 NCV1117ST33T3G
J28 USB_TYPE_C
J30
Y3
16MHz
D19 BUE05CN15
LPC55S69JEV98
L6 600OHM
L7 600OHM
L8 600OHM
L10 330OHM@100MHz
L9 4.7uH
LED_USB_COMM USB Communication
LED_SWO_ACT SWO activity
PIO0_5-ISP_EN-LED_SWD_ACT Status / SWD Activity
LED_VCOM_ACT VCOM Activity
FUNC Func
LED_SIO USB bridge active
LED_NRG Current measurement active
MCU_LINK_RST
VBAT_DCDC/VBAT_PMU
VREF_MCULINK
USB_DEBUG_VBUS
USB_MCU_LINK_DP
USB_MCU_LINK_DN
P5V_MCU_LINK_USB
SILK = DEBUG USB
SILK = LINK DBG
Jumper (Default) = 1-2
Jumper (Default) = OPEN
Open : Boot from Internal Flash
Closed: Enable ISP mode (USB1)
HW_VER_2: USB-SIO bridge disable when low
HW_VER_6: VCOM disable when low
HW_VER_7: SWD disable when low
ICAP Classification: CP: ___ IUO: X PUBI: _
Drawing Title: FRDM-MCXW70
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MCULink Debug Interface



Debug Trace Connector

FRDM-MCXW70

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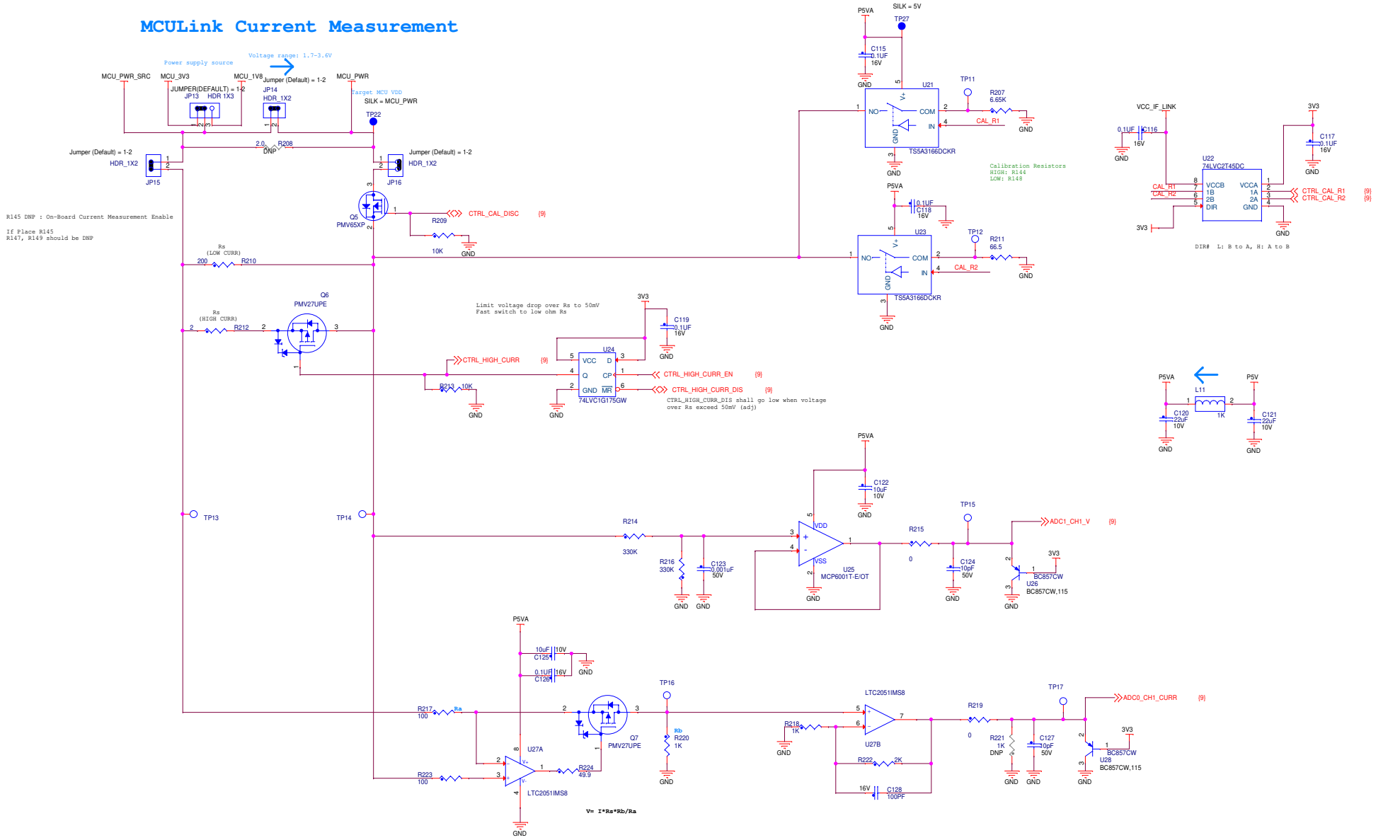
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Page Title: **MCU_LINK_DEBUG**

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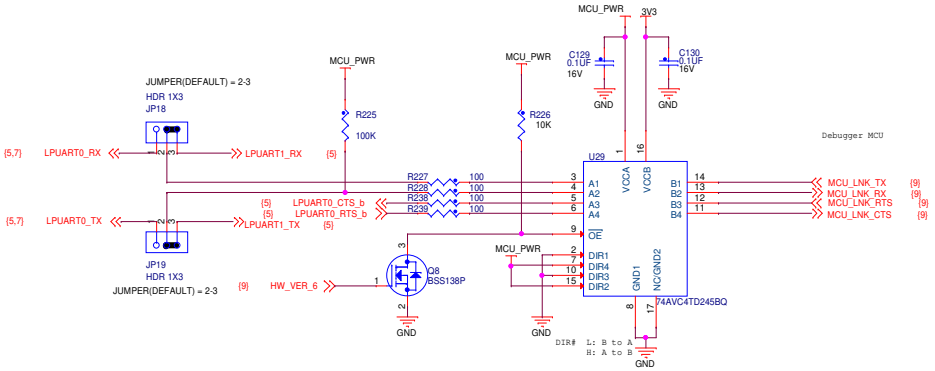
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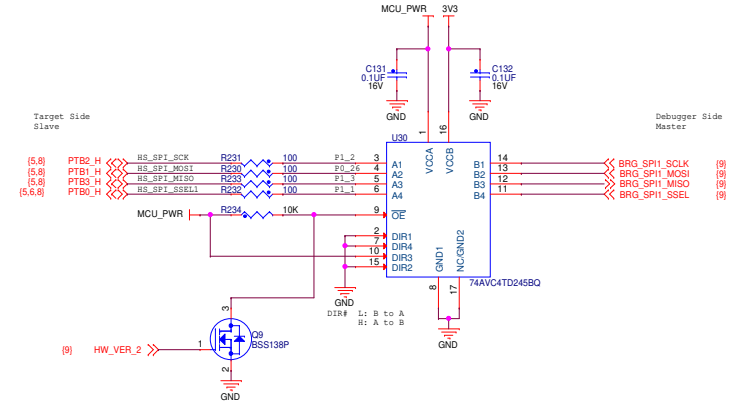
MCULink Current Measurement

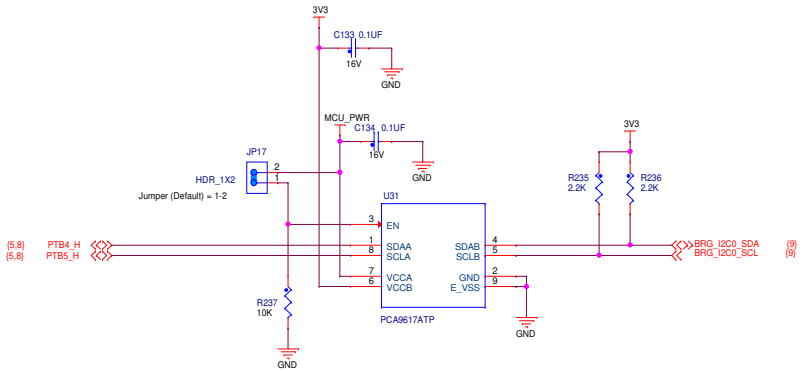


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Page Title:	MCU_LINK_CURRENT		
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MCULink Voltage Translation Buffers







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Drawing Title:

FRDM-MCXW70

Page Title:

MCU_LINK_VCOM_SIO

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